

DAY 9 -SUMMARY

1. OOP (Object-Oriented Programming)

A programming approach that uses classes and objects to structure software.

2. Abstraction

Hides implementation details and shows only the necessary features to the user.

3. Inheritance

Allows one class to acquire the properties and methods of another class, promoting code reusability.

4. Types of Inheritance

- Single Inheritance
- Multilevel Inheritance
- Hierarchical Inheritance
- Hybrid Inheritance (conceptual in Java)

5. Polymorphism

Allows the same method or object to perform different actions based on the context.

6. Types of Polymorphism

- Compile-Time Polymorphism (Method Overloading)
- Run-Time Polymorphism (Method Overriding)

7. Functions

Blocks of code designed to perform specific tasks and improve code reusability.

8. Class and Its Usage

A class is a blueprint for creating objects that contain data and methods.

Function Overloading

Defining multiple methods with the same name but different parameters within the same class.

Function Overriding

Redefining a parent class method in a child class to provide a specific implementation.

Multilevel Inheritance

A type of inheritance where a class inherits from another derived class, forming a chain of inheritance.